

Element I: Testing, Data Collection, and Analysis

Project Title: MJ's LAST Stand (Echo Rack)

1 - Importance of Tests

As a reminder (to ourselves and others) our five high priority requirements were the gong stand being able to hold the weight of the gong, the gong needs to be able to freely swing and stay balanced throughout, the gong stand can not be ugly, the gong stand should have an integrated mute, and the gong stand should have a relatively low cost. Further subsections will cover the plan for testing each individual requirement and why each requirement is important.

1. Hold Gong Weight:

- a. Although this may seem obvious, the gong stand needs to be able to hold the weight of the gong in order to function as a stand. If it can't hold the weight of the gong, then it can't do its sole purpose.
- b. We could test this requirement by simply putting the gong on the stand and seeing what happens in the next coming weeks, and see if we need to make any changes. This should be suitable for a short amount of time, but I'm not concerned about the gong holding a good amount of weight.
- c. Forget what I said in b, I'm just going to test it by putting my entire body weight on it. That's how much faith I have in our product. (It can hold roughly 140lbs) Unfortunately, I was not able to test this further because Mr. Paz didn't want to accidentally have a lawsuit if I got hurt. (more so he didn't want it to break right as it was about to be done)

2. Swing Freely and Balanced:

- a. The gong also needs to be able to swing back and forth. What this really means is that our project can't obstruct the movement of the gong. If the gong can't move and vibrate, then it can't make sound, and if it can't make sound, we've failed our project.
- b. This should be a really easy test, considering that the only thing we have to do is attach the gong to the stand and hit it a couple times with varying strength to see if anything we made would obstruct it.

3. Elegant Design

- a. The gong stand must look good and not be embarrassing to use at our concerts. This one is very important because we want it to be an improvement from our current gong stand. (although, I think anything would be an improvement at that point.)

- b. Despite our best efforts to make it look pretty, the ultimate test will just be going to Mr. Waller and asking him how it looks, and if he likes it. Obviously we like it, but we're also a little biased.

4. Integrated Mute

- a. Having an integrated mute means that when you do an action, in our case pushing a foot pedal, it can control the gong's vibrations, which means it can control if the gong rings or not.
- b. We can test this simply by hitting the gong, and making adjustments to the amount of pressure being applied based on how quickly the noise stops. More pressure theoretically means it stops ringing quicker.

5. Low Cost

- a. Yeah uhh... I'll be honest here, we were over budget. (which isn't unusual for me) Not sure what exactly we put here, because the project is already done and everything is already bought. This requirement was more for the planning process rather than the testing process.
- b. You just have to keep in mind that real companies buy in bulk and have specialized machines that can do more things for them at a cheaper price, so obviously anything we make would likely be more expensive. However, in this case, we were still able to undercut what other companies sell and we have more features. (more later)

2 - Results of Tests

1. Hold Gong Weight

- a. As previously mentioned, it can hold my weight so it's fine. (140 lbs) I could do pull ups on our gong stand (if I was strong enough to do pull ups). In more realistic terms, it was able to hold the weight of our gong via the hooks we had and survived our concert so we're all good.

2. Swing Freely and Balanced

- a. For the most part, successful. Occasionally, the lip of the gong will touch the arm of the mute, which isn't ideal, but it also doesn't do anything bad to either part. Other than that, it swings very well and still vibrates to make sound.

3. Elegant Design

- a. Very successful. Mr. Waller liked it so that's all that matters. The wood stain we used wound up matching the color of the brass of the gong, so

they looked very good together. Additionally, the blue was a nice touch and a good contrast color, especially given our school's color theme.

4. Integrated Mute

- a. Mostly successful. The mute is a little hard to use due to the foot pedal and the length of the strings, but it does mute the gong. Definitely the area that needs most improvement.

5. Low Cost

- a. The project was a *little* more money than we expected, but was still relatively cheaper compared to other good looking gong stands on the market. Obviously you can buy a little cheap metal thing for roughly \$100, but it doesn't come with locking wheels, an integrated mute, or any aesthetic design. We kind of failed this requirement, but also succeeded? Relatively?